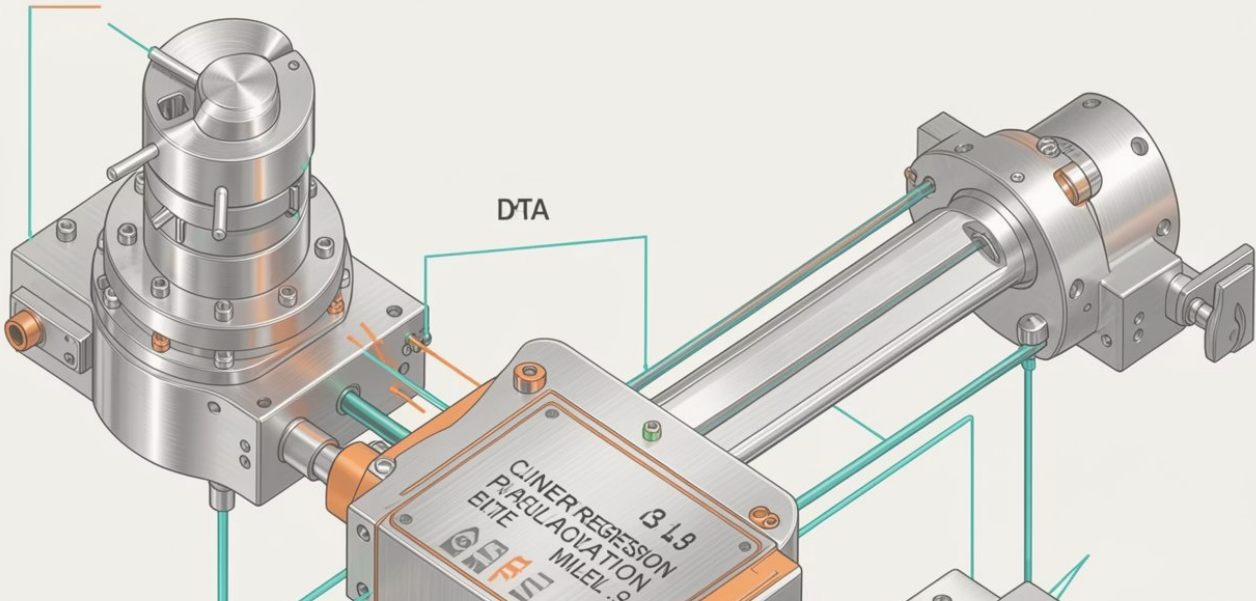


SALARY PREDICTION SYSTEM

Experience se accurate salary estimate

Document	SPS-SRS	Model	Linear Reg
Input	YearsExperience (Float)		
Output	Estimated Salary (Currency)		
Workflow	Data → Train → Predict → Viz		

Machine learning model jo professional experience ki base par reliable salary forecast return karta hai.



01 01 · Software Requirements Specification

02 02 · Linear Regression Foundation

03 03 · Data Ingestion & 30-Row Dataset

04 04 · Validation & Quality Gates

05 05 · Production UI & Real-time Serving

Experience input dein. Instant salary lein.

System ka primary flow simple hai: user experience enter karta hai, trained Linear Regression model quantified salary estimate return karta hai.

Primary Flow	Experience input se instant currency output
Model Engine	Supervised Linear Regression (Scikit-Learn)
Target Latency	Sub-second response (< 1 second)
Decision Impact	HR offer benchmark aur candidate expectation

STANDARD OPERATING PATH

- INPUT:** YearsExperience field
- VALIDATE:** Sanitization & range check
- PREDICT:** Fitted line execution
- RETURN:** Clear salary card output

INPUT EXCEPTION

- NEGATIVE EXP**
- NON-NUMERIC**
- OUT OF RANGE**

CONTRACT	Deterministic formula	No hidden variables	Explainable regression
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Made with GAMMA

Dataset chhota hai, lekin signal seedha hai

Requirements document 30 records report karta hai: 5.31 saal average experience par \$76,003 mean salary ka mazboot linear relationship hai.



DATASET · Salary_Data.csv
SAMPLE SIZE · 30 records

Field	Type	Meaning & Distribution
YearsExperience	Float	Total experience in years (Mean: 5.31 years)
Salary	Float	Yearly salary in USD (Mean: \$76,003)
Target Relation	Linear	Strong positive slope between experience aur compensation
Missing Values	Zero	Clean raw input baseline without imputation overhead

DATA PROFILE

30

Total rows in baseline

5.31 YRS

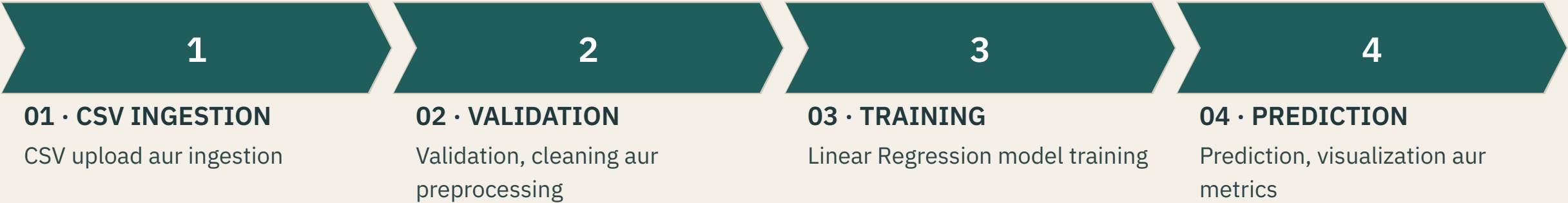
Mean YearsExperience

\$76,003

Mean annual salary

Data se prediction tak: four-step operating flow

Pipeline ko clear stages mein divide kiya gaya hai taake model repeatable aur explainable rahe.



 Data Ingest Salary_Data.csv parse karke Pandas DataFrame create hoti hai.	 Preprocess Null values check aur numeric float types verify kiye jaate hain.	 Fit Model Scikit-Learn se slope (m) aur intercept (c) calculate hotay hain.	 Predict & Plot User experience par salary compute hoti hai aur trendline draw hoti hai.
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Functional requirements product ko define karti hain

System 4 essential modules deliver karega taake end-to-end management, analytics aur real-time inference ek platform par hon.

REQ	MODULE	MUST DO (FUNCTIONAL CONTRACT)	DELIVERABLE		
01	Data Management	Upload, validate, clean; ensure correct format and missing value handling	Clean DataFrame		
02	Model Analytics	Train Linear Regression model; calculate summary statistics, R², MAE aur MSE	Model Artifact		
03	Prediction	Accept valid YearsExperience input; compute aur return formatted salary instantly	Salary Card Output		
04	Visualization	Render scatter plot with fitted regression line to visualize experience-salary trend	Interactive Plot		

STAKEHOLDERS

Teen user groups, teen practical needs

SPS ka value har user ke liye mukhtalif hai, lekin sub ek hi decision point par align hotay hain: rational salary estimate.

CORE METRIC

All roles use identical mathematical regression baseline.

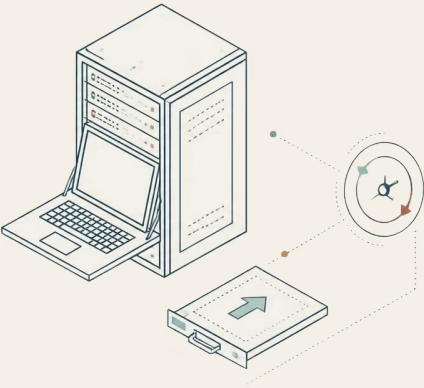


HR / RECRUITER

Offer Benchmarking

Candidate experience ke mutabiq fair aur competitive job offer determine karna taake hiring budget aligned rahe.

Goal: Data-backed compensation bands

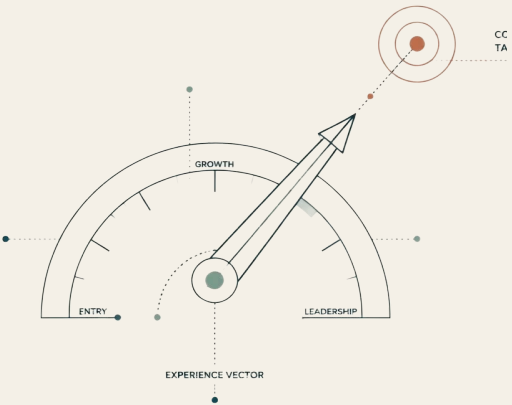


ADMIN / DATA ENG

Dataset & Retraining

Naya salary dataset upload karna, cleaning pipeline supervise karna aur updated model ko serialize karke serve karna.

Goal: Accurate model versioning & audit



JOB CANDIDATE

Market Expectation

Apne total YearsExperience ko enter karke market-rate salary benchmark jaan-na taake negotiations transparent hon.

Goal: Realistic, objective earning benchmark

Architecture: lightweight stack, clear boundaries

Recommended implementation Python-based backend aur simple responsive interface ko combine karti hai.

Layer	Suggested technologies
Data / model	Pandas, NumPy, Scikit-Learn
Backend	Flask ya FastAPI
Frontend	HTML, CSS, JavaScript, React ya Streamlit
Storage	CSV ya PostgreSQL

Prediction experience: input se salary card tak

UI ko intentionally simple rakha gaya hai taake user ko sirf required decision path nazar aaye.

STEP 1 · USER INPUT

YearsExperience

Text box ya slider.

ILLUSTRATIVE INPUT

3.5

Years of experience entered

Predict Salary

STEP 2 · MODEL INFERENCE

OUTPUT CARD

Predicted Annual Salary

\$60,500

Illustrative calculation: 3.5 years experience

STEP 3 · CONTEXT

Optional chart

Context ke liye visualization.

Quality bar: fast, accurate, safe

Non-functional requirements system ko real-world usable product banati hain, sirf theoretical prototype nahi.

Dimension	SRS Requirement	Acceptance Target	
Performance	Prediction result within 1 second	< 1,000 ms response time	
Accuracy	Target $R^2 \geq 0.90$ for high fidelity fit	R^2 metric calculation on test split	
Security	Sanitized inputs and secure API handling	Reject non-numeric & malicious payloads	
Usability	Mobile-responsive UI and clear error messages	Cross-device accessibility	
Reliability	99.9% uptime and serialized model persistence	Pickle / Joblib versioned storage	

QUALITY GATES

Acceptance Criteria

Latency < 1.0s
Strict API SLA

$R^2 \geq 0.90$
Explained variance

Zero Injection
Strict schema types

Har release in 5 non-functional dimensions par verify hogi.

Launch checklist: from SRS to usable SPS

Implementation ko structured stages mein execute karein taake data pipeline aur model behavior dono production-ready hon.

STAGED EXECUTION STEPS

01

Dataset Ingestion & Validation

Salary_Data.csv parse karke schema, missing values aur float conversions check karein.

02

Baseline Linear Regression Model

Scikit-Learn se train karein aur slope aur intercept parameters lock karein.

03

Calculate R², MAE & MSE Metrics

Verify karein ke model R² target (≥ 0.90) aur error boundaries meet karta hai.

04

Scatter Plot & Fitted Trendline

Data points aur regression trajectory visualize karke UI mein integrate karein.

05

Responsive Prediction UI & API

FastAPI / Flask endpoints ke saath Streamlit ya React interface connect karein.

06

Security, Serialization & Monitoring

Input validation sanitization apply karein aur Joblib se model persistent banayein.

DEPLOYMENT GATES

- ✓ Dataset ingestion complete
- ✓ Linear regression trained
- ✓ $R^2 \geq 0.90$ validated
- ✓ Trendline plot verified
- ✓ UI mobile responsive
- ✓ Model serialized safely

READY FOR DEPLOYMENT

CLOSING PRINCIPLE

Simple interface ke peeche disciplined data pipeline aur measurable quality gates lazmi hain.